

Andy Ta

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EDUCATION

University of Texas at Arlington

Bachelor of Science in Computer Science; GPA: 4.0/4.0

Arlington, TX

Expected May 2028

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C, HTML/CSS, Alloy

Frameworks/Libraries: React, React Native, Spring Boot, FastAPI, Node.js, Express.js, Tailwind CSS, Blockly

Tools/Databases: Git, GitHub, Jira, PostgreSQL, Supabase, Docker, k6, pytest, Jest, Postman, LabVIEW

Development: Full-Stack Development, REST APIs, Object-Oriented Programming, Unit Testing, Integration Testing, Agile/Scrum

EXPERIENCE

Test Engineer Intern

Lockheed Martin Missiles and Fire Control

June 2026 – Present

Grand Prairie, TX

- Develop a **LabVIEW GUI** integrating multiple power supplies through a Hardware Abstraction Layer (HAL) for simulation, verification, and test-data collection.
- Apply **object-oriented design** and reusable HAL interfaces to separate application logic from device-specific behavior across laboratory integrations.
- Collaborate with test, systems, and electrical engineers to debug software-hardware integrations and adapt software to evolving technical requirements.

Undergraduate Researcher

University of Texas at Arlington (SCOPE Lab)

February 2026 – Present

Arlington, TX

- Research block-based interfaces for the **Alloy formal modeling language**, comparing alternative architectures and evaluating usability and information-flow trade-offs.
- Build **30+ custom Blockly components** and React code generators that transform visual models into executable Alloy specifications.
- Develop an end-to-end execution pipeline with **Spring Boot and the Alloy Java API**, executing generated models and returning structured solver data for interactive React Flow visualizations.

PROJECTS

Project Lou – HackRICE | *Python, FastAPI, Docker, PostgreSQL, pytest, k6, Gemini*

September 2026

- Built a software maintenance tool that compares working and updated code to detect performance regressions that regular unit tests may miss.
- Created repeatable **Docker-based test environments** using pytest and k6 to compare baseline, changed, and fixed versions under the same conditions.
- Implemented independent fix verification with fresh Git worktrees, timeout and failure handling, and evidence-based checks before accepting proposed changes.

The bugHouse | *React, TypeScript, FastAPI, Supabase, PostgreSQL, Python*

March 2026 – Present

- Develop a full-stack attendance platform for UTA's Computer Science tutoring center, replacing paper sign-in with a **magnetic stripe ID scanning workflow**.
- Build responsive **React and TypeScript** dashboards with role-based views, reusable components, and real-time attendance analytics backed by FastAPI and PostgreSQL.
- Improve frontend responsiveness by storing frequently used aggregate values in PostgreSQL, reducing repeated client-side calculations on a **7-member Agile team**.

Fluuirish – FidHacks 1st Place | *React Native, TypeScript, FastAPI, PostgreSQL, Docker*

July 2026

- Built a gamified financial literacy application that won **1st place among 11 teams** at Fidelity Investments' FidHacks Westlake 2026.
- Developed user-facing features with React Native and TypeScript, integrating **FastAPI REST APIs and PostgreSQL** for learning data and personalized experiences.
- Integrated the **Gemini API** with curated financial content to generate domain-specific quizzes while contributing budgeting and stock-market features.

ACTIVITIES & LEADERSHIP

Association for Computing Machinery (ACM) @ UTA

Community Officer & ACM Create Member

August 2025 – Present

Arlington, TX

- Plan and execute technical and social events for **100+ students**, coordinating logistics, communications, scheduling, and new-member outreach.